

LESSON 6-3 · SESSION 2 · PRACTICE · FORM A**Finding the Value of a Power**

Learning target – I can find the value of a power by multiplying the base the number of times the exponent shows. · 6.EE.A.1

REMEMBER

- Write the **expanded form**, then multiply one factor at a time.
- $4^3 = 4 \times 4 \times 4 = 64$, **not** $4 \times 3 = 12$.
- Decimals and fractions work the same way: $0.3^2 = 0.3 \times 0.3 = 0.09$.

WORKED EXAMPLE

Find the value of 3^4 .

Step 1 Expand: $3^4 = 3 \times 3 \times 3 \times 3$.

Step 2 Multiply: $3 \times 3 = 9$, $9 \times 3 = 27$, $27 \times 3 = 81$.

Step 3 Check: I used 3 as a factor 4 times.

Answer $3^4 = 81$

WORDS evaluate (*evaluar*) · squared (*al cuadrado*) · cubed (*al cubo*) · expanded form (*forma desarrollada*)

PRACTICE

show your work · label units

1. Find the value of each power.

a. $12^2 =$ _____

b. $3^3 =$ _____

c. $2^4 =$ _____

d. $11^2 =$ _____

e. $1.2^2 =$ _____

f. $(1/3)^2 =$ _____

2. **Power Table.** A science club makes a table of powers of 2. Complete it. Then tell what each value is multiplied by to get the next one.

Power	Expanded form	Value
2^1		
2^2		
2^3		
2^4		

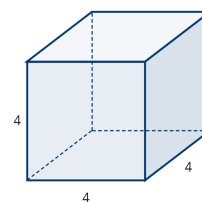
Multiply by _____

3. **Storage Cube.** A storage cube has edges of 4 units, so its volume is 4^3 cubic units. Find the volume.

Hint Use 4 as a factor three times.

Show your work.

Answer _____ cubic units



4. **Plant Cuttings.** A count of plant cuttings is multiplied by 5 each week for 4 weeks, so it is 5^4 . Evaluate it.

Hint $5 \times 5 = 25$, then keep multiplying by 5.

Show your work.

Value _____

5. **Poster Claim.** A classroom poster says $7^2 = 14$. Justify or correct the poster.

Hint Expand the power into factors first.

Show your work.

Value _____

6. **Which Is Biggest?** Evaluate 2^5 , 5^2 and 3^3 . Order them from least to greatest.

Hint Evaluate each power before comparing.

Show your work.

Order _____

7. **Decimal Powers.** Evaluate each power.

Hint Count decimal places: 0.6×0.6 has two.

a. $0.6^2 =$ _____

b. $1.5^2 =$ _____

8. **Growing Cells.** A culture of cells triples every hour. It starts with 1 cell.

a. Write the number of cells after 4 hours as a power of 3, then evaluate. _____ cells

b. How many cells after 5 hours? _____ cells

c. How many more cells after 5 hours than after 4 hours? _____ cells

FIND THE ERROR

explain the mistake, then fix it

9. **Teo's Work.** Teo evaluated 10^3 . He wrote $10 \times 3 = 30$ and said the value is 30. Explain his mistake. Then find the correct value.

TEST PRACTICE

choose the best answer

10. 2^5 equals –

A 10

B 25

C 32

D 7

11. A number to the 2nd power is said to be –

A squared

B cubed

C doubled

D halved

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

2.25 R	64 S	81 E	0.36 A	162 D	20 X	49 U	12 Z	14 J	625 Q
#3	#4	#5	#7a	#7b	#8a	#8c			

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____